

Kira DeVore

Ph.D. Candidate, School of Molecular Sciences, Biodesign Center for Applied Structural Discovery

Arizona State University Biodesign Building C377, 1001 S. McAllister Ave, Tempe, AZ 85287

kdevore1@asu.edu | [GitHub](#) | [LinkedIn](#) | [ResearchGate](#) | [ORCID](#)

Education

Ph.D., Biochemistry 2022 – Present

Arizona State University, Tempe, AZ

School of Molecular Sciences | Biodesign Center for Applied Structural Discovery

Advisor: Po-Lin Chiu

B.Sc., Biochemistry (Medicinal Chemistry) 2018 – 2022

Arizona State University, Tempe, AZ

School of Molecular Sciences

GPA: 4.0

B.Sc., Biological Sciences (Genetics, Cell, and Developmental Biology) 2018 – 2022

Arizona State University, Tempe, AZ

School of Life Sciences

GPA: 4.0

Publications

Published:

Chen, L., Li, Z., Yan, J., Dey, S., Li, C., Petrova, A., Prasad, A., Satyabola, D., **DeVore, K.**, Tu, X., Qu, Y., Wang, S., Wu, G., Chiu, P.-L., Wen, D., Wang, J., Yan, H., Yin, P., Liu, D. Programmable Self-Assembly of RNA Nanostructures with >100 Unique Components. *bioRxiv* <https://doi.org/10.64898/2026.09.02.748990> (2026).

Nandi, P. †, **DeVore, K.** †, Wang, F. †, Li, S., Walker, J. D., Truong, T.T., LaPorte, M.G., Wipf, P., Schlager, H., McCleerey, J., Paquette, W., Columbres, R.C.A., Gan, T., Poh, Y.-P., Fromme, P., Flint, A.J., Wolf, M., Huryn, D.M., Chou, T-F., Chiu, P.-L. Mechanism of the allosteric inhibition of triazole class inhibitors on human p97/VCP ATPase and its disease mutants by a class of triazole inhibitors. *Nature Communications Chemistry* 7:177 (2024).

DeVore, K. and Chiu, P.-L. Probing structural perturbation of biomolecules by extracting cryo-EM data heterogeneity. *Biomolecules* 12:628 (2022).

Under Review:

DeVore, K. †, Lee, Y. †, Kitchel, H. J. †, Gorski, C., Dadwal, A., Ferrer, E., Tovmasyan, D., Poh, Y-P., Mazor, Y., Liu, H., Chiu, P.-L. Integrated multiscale profiling reveals threshold-like photoacclimation in *Chlorobaculum tepidum*, a model green sulfur bacterium. Photosynthesis Research [In Review].

Presentations

Oral presentation:

Annual Cyanobacteria and Algae Research and Education (CARE), Alabama State University, Virtual (2025)

“Structural Insights into the Rubisco-Rubisco Activase Complex in Higher Order Plants”

International Conference on Crystallization of Biological Macromolecules, Arizona State University, Tempe, AZ (2024)

“Structural study of human neurotrophin receptor complex implicated in neuronal apoptosis and neurodegeneration”

Poster presentation:

DeVore, K., Lee, Y., Kitchel, H. J., Gorski, C., Dadwal, A., Ferrer, E., Tovmasyan, D., Poh, Y.-P., Mazor, Y., Liu, H., Chiu, P.-L. (2026) Multi-scale mechanisms of light adaptation in green sulfur bacterium *Chlorobaculum tepidum*. 43rd American Society for Photobiology Biennial Meeting, Scottsdale, AZ

DeVore, K., Waheeda, K., Breen, I., Wachter, R., Chiu, P.-L. (2026) Structural Insights into the Rubisco-Rubisco Activase Complex in Higher Order Plants. School of Molecular Sciences Research Symposium, Tempe, AZ

DeVore, K., Waheeda, K., Breen, I., Wachter, R., Chiu, P.-L. (2026) Structural Insights into the Rubisco-Rubisco Activase Complex in Higher Plants. Arizona Imaging and Microanalysis Society, Tempe, AZ

DeVore, K., Waheeda, K., Breen, I., Wachter, R., Chiu, P.-L. (2026) Structural Insights into the Rubisco-Rubisco Activase Complex in Higher Order Plants. Biodesign Fusion Retreat, Tempe, AZ

DeVore, K., Waheeda, K., Breen, I., Wachter, R., Chiu, P.-L. (2026) Structural Insights into the Rubisco-Rubisco Activase Complex in Higher Order Plants. Biophysical Society Conference, San Francisco, CA

DeVore, K., Nandi, P., Kitchel, H., Poh, Y-P., Chou, T-F., Chiu, P.-L. (2025) Structural study of human neurotrophin receptor complex implicated in neuronal apoptosis and neurodegeneration. Biophysical Society Conference, Los Angeles, CA

DeVore, K., Nandi, P., Kitchel, H., Poh, Y-P., Chou, T-F., Chiu, P.-L. (2024) Structural study of human neurotrophin receptor complex implicated in neuronal apoptosis and neurodegeneration. International Conference on Crystallization of Biological Macromolecules, Tempe, AZ

DeVore, K., Nandi, P., Poh, Y-P., Chiu, P.-L. (2023) Structural study on the proNGF-p75^{NTR}-sortilin receptor complex implicated in neuronal apoptosis. Arizona Alzheimer's Consortium, Tempe, AZ

DeVore, K., Kohli, A., Nisanov, A. (2022) Investigating the role of lncRNA and protein structure in differential regenerative capacities between *Homo sapiens* & *Anolis carolinensis*. Arizona State University, Tempe, AZ

DeVore, K. (2020) The Effect of Substituent Groups on Opiate Receptor Binding. Arizona State University, Tempe, AZ

Experience

Research

Center for Applied Structural Discovery, Biodesign C 2021 – Present

School of Molecular Sciences, Arizona State University, Tempe, AZ

Advisor: Po-Lin Chiu, Ph.D.

- Research primarily focuses on structural biology and cryo-EM. Individual focuses include structural and functional characterization of protein complexes related to photosynthesis as well as cryo-EM method development.

Center for Single Molecule Biophysics, Biodesign B 2018 – 2020

School of Molecular Sciences, Arizona State University, Tempe, AZ

Advisor: Marcia Levitus, Ph.D.

- Research primarily focuses on understanding DNA dynamics and protein oligomerization using fluorescence spectroscopy and protein-induced fluorescence enhancement (PIFE). Individual focus on uracil-DNA glycosylase efficiency and how it is modulated by substrate rigidity.

Teaching

Graduate Teaching Assistant 2022 – Present

School of Molecular Sciences, Arizona State University, Tempe, AZ

- Instructed groups of students in undergraduate-level biochemistry laboratory courses – experiments including primer design, PCR, cloning, transformation, *E. coli* cell culture & overexpression, affinity chromatography, ion exchange chromatography, SDS-PAGE, protein concentration assays, optical property determination, and enzyme kinetics

Professional

Tutor 2018 – Present

- Instructed and tutored students in high-school and college-level courses within Chemistry, Biology, Physics, Mathematics, and English. Delivered individualized instruction to students to address their specific needs and learning strategies

Mathematics Instructional Aide

2019 – 2022

- Instructed and tutored students in college-level mathematics courses (College Algebra, College Mathematics, and Calculus I w/ Analytical Geometry)

Mathematics Grader

2019 – 2022

- Provided neat and timely grading for students on assignments. Utilized tools, such as Microsoft Office, Microsoft Excel, Canvas, Blackboard, SpeedGrader, Google Docs, and Google Sheets

Honors and Awards

School of Molecular Sciences TA Appreciation Award

June 2026

School of Molecular Sciences, Arizona State University, Tempe, AZ

ASP Photosynthesis and Photoconversion Division Poster Award

May 2026

Amount: \$150

American Society for Photobiology, Scottsdale, AZ

Outstanding Graduate Research Poster Award

April 2026

Amount: \$1,000

School of Molecular Sciences, Arizona State University, Tempe, AZ

1st Place BioEM Poster Award

April 2026

Amount: \$150

Arizona Imaging and Microanalysis Society, Tempe, AZ

Swift Scholarship

August 2025

Amount: \$2,250

Arizona State University, Tempe, AZ

The College Graduate Excellence Award Amount: \$100 Arizona State University, Tempe, AZ	April 2025
69th Annual Meeting Travel Grant Amount: \$250 Biophysical Society, Los Angeles, CA	February 2025
School of Molecular Sciences University Graduate Fellowship Amount: \$300 Arizona State University, Tempe, AZ	January 2025
School of Molecular Sciences Outstanding Graduate Student Merit Award Amount: \$1,500 Arizona State University, Tempe, AZ	September 2024
Teaching Excellence Award Amount: \$750 Arizona State University, Tempe, AZ	September 2024
Outstanding Mentor Award Amount: \$750 Arizona State University, Tempe, AZ	September 2024
Swift Scholarship Amount: \$2,250 Arizona State University, Tempe, AZ	August 2024
Outstanding Teaching Assistant Award	April 2024

Arizona State University, Tempe, AZ

Distinguished Biochemistry Merit Award

May 2022

Arizona State University, Tempe, AZ

Dean's List Recipient (x8)

2018 – 2022

Arizona State University, Tempe, AZ

New American University Scholarship

2018 – 2022

Amount: \$32,000

Arizona State University, Tempe, AZ

Certifications

Graduate Student Research Responsible Conduct of Research (RCR)

June 2024

CITI Program

Credential ID: 63277257

Certified Entry-Level Python Programmer (PCEP)

March 2024

Open EDG Python Institute

Credential ID: fN7B.54LQ.PMEJ

Computational Life Sciences

2018 – 2022

Arizona State University, Tempe, AZ

Evolutionary Medicine

2018 – 2022

Arizona State University, Tempe, AZ

Technical Skills

Programming Languages: Python, R, MATLAB, HTML, and CSS

Software: cryoSPARC, Relion, Coot, Chimera, ChimeraX, Tomography 5, EPU, Microsoft Excel, Microsoft Word, Microsoft PowerPoint, Microsoft Excel, SnapGene, SerialEM, Benchling, Etomo, and ChemDraw

Laboratory Skills: agarose gel electrophoresis, SDS-PAGE, BN-PAGE, western blotting, DNA transformation and isolation, primer design and PCR, cloning, *Sf9* cell culture, baculovirus *Sf9* expression, *E. coli* expression, protein purification, ion exchange chromatography, affinity chromatography, size-exclusion chromatography (FPLC), pigment extraction, high performance liquid chromatography (HPLC), negative stain EM, TEM operation, cryo-EM, EM image processing, fluorescence spectroscopy, UV-Vis spectroscopy, protein concentration assays, ultracentrifugation, plunge-freezing, mass photometry, RNA-seq, circular dichroism (CD), dynamic light scattering (DLS), and fluorescence microscopy

Outreach and Media Coverage

Image Featured in BPS AZ Chapter Calendar	April 2025
ASU Visitation Weekend	March 2025
School of Molecular Sciences celebrates high-achieving grad students	January 2025
High School Student STEM Mentor	October 2024
Biodesign Fall Town Hall Recognition	September 2024
Nature Communications Chemistry 'Behind the Paper' Contribution	August 2024
BioSense Network	July 2024
Biodesign Open Door	March 2024
Academic Liaison	April 2023
Biodesign Open Door	March 2023

References

Po-Lin Chiu, Associate Professor in School of Molecular Sciences, Arizona State University, plchiu@asu.edu, 480-727-2058

Marcia Levitus, Associate Professor in School of Molecular Sciences, Former Associate Director of Graduate Programs, Arizona State University, mlevitus@asu.edu, 480-727-8586

Scott Lefler, Teaching Professor in School of Molecular Sciences, Arizona State University,
slefler@asu.edu, 480-727-8282